
IT3010 NDM
SNMP
SNMP
SNMP
MID past paper
MCQ EXPLANATION

Select the correct statement regarding the process of configuration management, under ISO framework for network management.

Select one:

- a. Gather, modify and store configuration information
- b. Gather, manipulate and store configuration information
- c. Gather, store and process configuration information
- d. Gather, store and distribute configuration information

Correct answer: a. Gather, modify and store configuration information

In the **ISO framework for network management**, **Configuration Management** consists of:

- **Gathering** information about the current network environment
- **Modifying** the configuration of network devices using that data
- **Storing** the data and maintaining an up-to-date inventory/report system

Lecture 1 / Introduction to Network Design & Management

Topic: **ISO Network Management Framework → Configuration Management**

Recommend when to start baselining process?

Select one:

- a. When there is high congestion
- b. In special occasions
- c. From the start point of the network functioning
- d. When system administrator feels that the network speed is low

Correct answer: c. From the start point of the network functioning

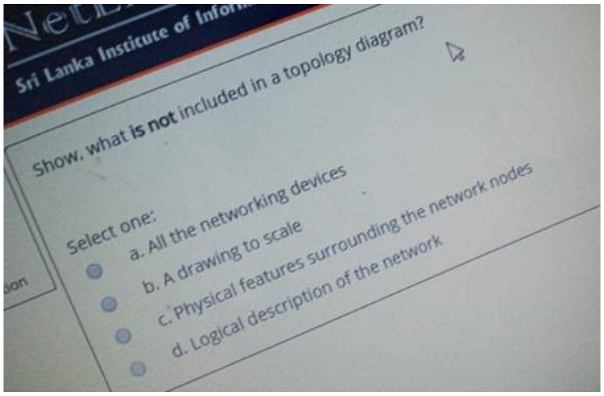
The lecture on **Baselining** clearly says:

- **"Begin immediately"**
- Baselining should be **long term** and done from the beginning of network operation

Because a baseline is supposed to capture the **normal operating condition** of the network. If you only start later, after problems appear, you miss the true normal behavior.

Lecture 2 – Network Servers, Baselining & Auditing

Topic: **When to Baseline**



Correct answer: b. A drawing to scale

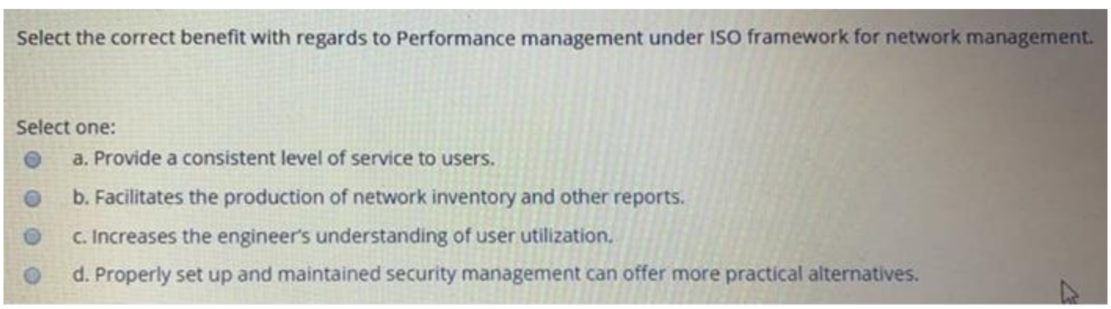
A **topology diagram** shows the **logical arrangement of the network**:

- devices
- connections between devices
- logical structure of the network

It is **not required to be drawn to scale**.

A scale drawing belongs more to **topography/floor plan / physical location documentation**, not a topology diagram.

Lecture 2 – Network Servers, Baselining & Auditing
Topic: **Network Mapping**



Correct answer: a. Provide a consistent level of service to users

Performance Management is described as helping reduce congestion and inaccessibility so the network can **provide a consistent level of service to users**.

Lecture 1 / Introduction to Network Design & Management
Topic: **ISO Network Management Framework (FCAPS)**

Identify the correct devices which transmits timed broadcasts onto an active network?

Select one:
- a. DHCP client
- b. Unmanaged switch
- c. Firewall
- d. Bridge

Correct answer: a. DHCP client

A **DHCP client** periodically sends **broadcast messages** on the network to obtain or renew an IP address. Examples of DHCP broadcast messages:

- **DHCPDISCOVER**
- **DHCPREQUEST**

These are broadcast packets sent to: 255.255.255.255

They are **timed broadcasts** because DHCP clients periodically:

- renew leases / search for a DHCP server / rebind addresses

So the device that **transmits timed broadcasts** is the **DHCP client**.

Lecture 2 – Network Servers, Baselining & Auditing

Topic related to: **Network traffic / Broadcast traffic /Devices generating network activity**

Discover the correct statement with respect "ROI" (Return On Investment) related to Network Managment?

Select one:
- a. Any component of a system that upon failure will cause a malfunction in the entire system
- b. The average time a device will function before failing
- c. Is the average time before an electronic component can be expected to require repair
- d. An accounting formula used to obtain an actual or perceived future value of an expense

Correct answer: d. An accounting formula used to obtain an actual or perceived future value of an expense

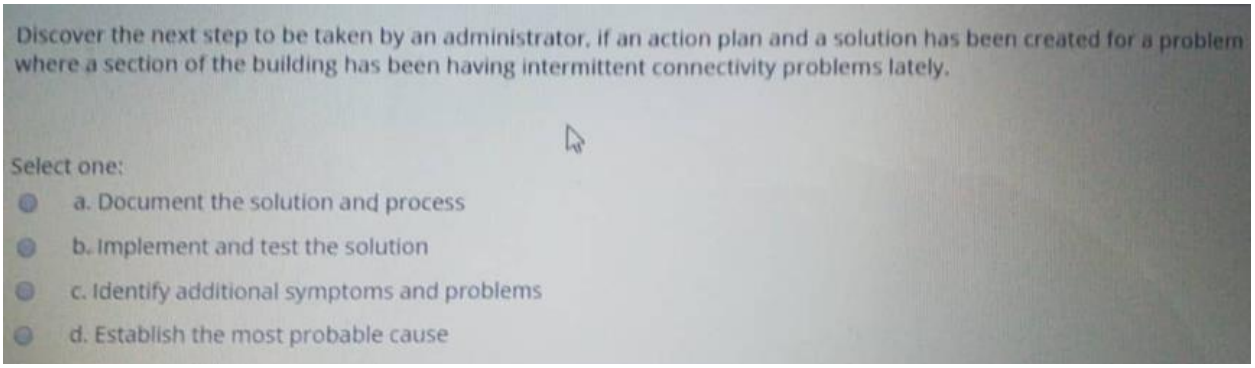
ROI (Return On Investment) is defined as: an accounting formula used to obtain an actual or perceived future value of an expense or investment.

So option **d** matches the lecture definition directly.

Lecture 2 – Network Servers, Baselining & Auditing

Topic: **Glossary / Vocabulary**

- **ROI** → investment value
- **SPOF** → one failure breaks system
- **MTBF** → how long before failure
- **MTTR** → repair-related time



Discover the next step to be taken by an administrator, if an action plan and a solution has been created for a problem where a section of the building has been having intermittent connectivity problems lately.

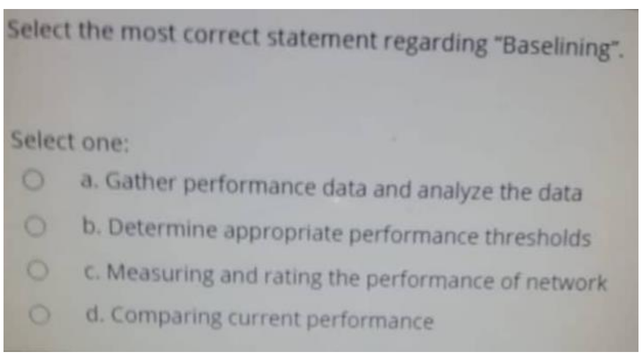
Select one:
- a. Document the solution and process
- b. Implement and test the solution
- c. Identify additional symptoms and problems
- d. Establish the most probable cause

Correct answer: b. Implement and test the solution

Use this sequence:

1. Identify symptoms
2. Find probable cause
3. Create an action plan
4. Implement and test
5. Document

Lecture 1 – Fault Management / Network Management Process
also overlaps with **Network Monitoring / Troubleshooting logic**



Select the most correct statement regarding "Baselining".

Select one:
- a. Gather performance data and analyze the data
- b. Determine appropriate performance thresholds
- c. Measuring and rating the performance of network
- d. Comparing current performance

Correct answer: c. Measuring and rating the performance of network

Network baselining as: the act of measuring and rating the performance of a network in real-time situations

Lecture 2 – Network Servers, Baselining & Auditing

Topic: **Baselining / Network Baselining**

Compute the correct implementation to take place if, you are the network administrator of ABC_Computers and there is a requirement to reduce the bandwidth used between the ABC_Computers internal network and the Internet.

Select one:

- a. Proxy server
- b. HTTP server
- c. WINS (Windows Internet Name Service) server
- d. DHCP server

Correct answer: Proxy server

The requirement is to **reduce bandwidth use between the internal network and the Internet**. A **proxy server** is the best implementation because it can **cache frequently requested web content**. When multiple users request the same website or file, the proxy can serve it from its cache instead of downloading it again from the Internet.

Lecture 2 – Network Servers, Baselining & Auditing

Topic: **Proxy Server / Proxy Cache**

What is network congestion?

Select one:

- a. How much time it takes for a data packet to travel from one designated point to another.
- b. The amount of material or items passing through a system or process.
- c. The reduced quality of service that occurs when a network node or link is carrying more data than it can handle.
- d. The amount of time it takes for the head of the signal to travel from the sender to the receiver.

Correct answer: c. The reduced quality of service that occurs when a network node or link is carrying more data than it can handle.

Network congestion happens when the traffic load becomes higher than what the network node or link can handle properly.

As a result:

- delays increase
- packets may be dropped
- service quality becomes poor

T**Lecture 1 – Performance Management**

The lecture lists **congestion** as one of the important performance terms.

Lecture 4 – Network Monitoring

Also related to monitoring utilization and overloaded links.

Select the true statement regarding TCP and UDP?

Select one:
- a. TCP is connection-oriented, UDP is unreliable
- b. TCP is connectionless, UDP is reliable
- c. TCP is connectionless, UDP is unreliable
- d. TCP is connection-oriented UDP is reliable

Correct answer: a. TCP is connection-oriented, UDP is unreliable

TCP = Trustworthy Connection Protocol **UDP** = Unreliable Datagram Protocol

TCP = connection-oriented and reliable **UDP =** connectionless and unreliable

Feature	TCP	UDP
Full Name	Transmission Control Protocol	User Datagram Protocol
Connection Type	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Data Delivery	Guarantees delivery	No guarantee of delivery
Order of Packets	Packets arrive in correct order	Packets may arrive out of order
Error Handling	Error detection + recovery	Error detection only, no recovery
Acknowledgement	Uses acknowledgements (ACK)	No acknowledgements
Retransmission	Lost packets are retransmitted	Lost packets are not retransmitted
Speed	Slower	Faster
Overhead	Higher overhead	Lower overhead
Flow Control	Yes	No
Congestion Control	Yes	No
Header Size	Larger	Smaller
Best Use Cases	Web, email, file transfer	Streaming, gaming, DNS, VoIP
Examples	HTTP, HTTPS, FTP, SMTP	DNS, DHCP, VoIP, online gaming

Compare the major disadvantage of a hierarchical network management architecture over distributed network management architecture?

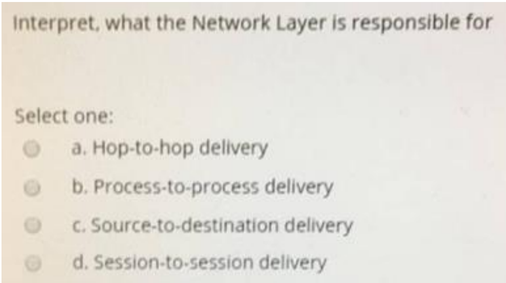
Select one:
- a. Distribution of network management tasks
- b. Centralized information storage
- c. Combines centralized and hierarchical approaches
- d. Distributed monitoring

Correct answer: centralized information storage

In a **hierarchical network management architecture**, information and control are still arranged in levels, and this often leads to **centralized storage of management information** at upper levels.

Lecture 1 – Introduction to Network Design & Management

Topic: **Network Management Architecture**


Interpret, what the Network Layer is responsible for

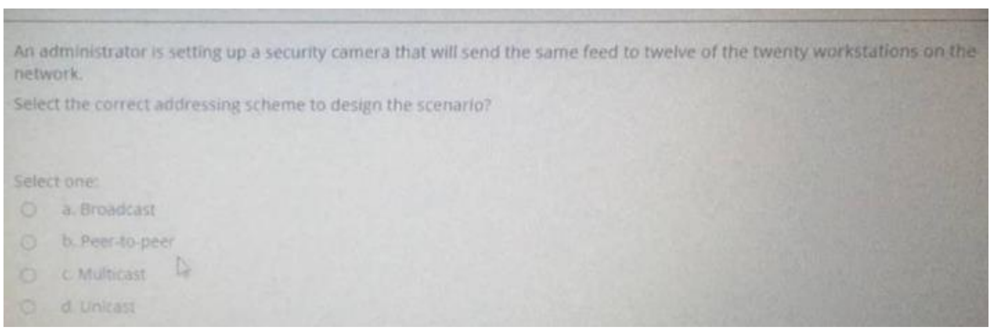
Select one:
a. Hop-to-hop delivery
b. Process-to-process delivery
c. Source-to-destination delivery
d. Session-to-session delivery

Correct answer: source-to-destination delivery

The **Network Layer** is responsible for moving packets from the **source host to the final destination host** across one or more networks.

Layer	Responsibility
Data Link	Hop-to-hop
Network	Source-to-destination
Transport	Process-to-process

Lecture 2 – Network Servers, Baselining & Auditing
Topic: **OSI Model / Network Layer**


An administrator is setting up a security camera that will send the same feed to twelve of the twenty workstations on the network.
Select the correct addressing scheme to design the scenario?

Select one:
a. Broadcast
b. Peer-to-peer
c. Multicast
d. Unicast

Correct answer: multicast

The camera needs to send the **same feed** to **12 specific workstations out of 20**, not to everyone.
That is exactly what **multicast** is for:

- **one sender**
- **selected group of receivers**
- efficient delivery to multiple intended devices

Lecture 4 – Network Monitoring
Topic: **Broadcasts / Multicasts / network traffic types**

Interpret the correct service/s used by DNS using the well-known port 53.

Select one:
- a. UDP
- b. SMTP and UDP
- c. TCP
- d. Either TCP or UDP

Correct answer: either TCP or UDP

DNS (Domain Name System) uses **port 53**, but it can operate using **both UDP and TCP** depending on the situation.

Service	Port	Protocol
DNS	53	UDP / TCP
HTTP	80	TCP
HTTPS	443	TCP
SMTP	25	TCP
DHCP	67 / 68	UDP
SNMP	161	UDP

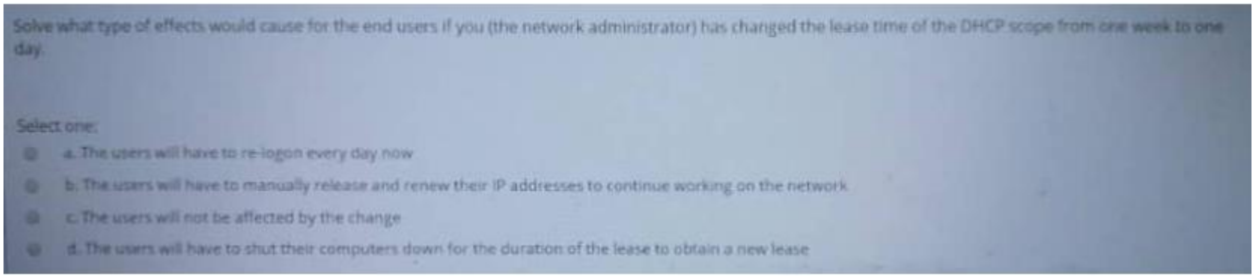
Lecture 2 – Network Servers, Baselining & Auditing
Topic: **DNS server / network services**

Outline the correct definition of Network Management?

Select one:
- a. The process of controlling complex data networks to maximize its efficiency and productivity.
- b. The process of controlling complex data networks to maximize its scalability and manageability.
- c. The standard of controlling data networks to maximize its agility and productivity.
- d. The standard of controlling data networks to maximize its suitability and integrity

Correct answer:The process of controlling complex data networks to maximize its efficiency and productivity

Lecture 1 – Introduction to Network Design & Management
Topic: **The Network Management Process**


Solve what type of effects would cause for the end users if you (the network administrator) has changed the lease time of the DHCP scope from one week to one day.

Select one:
a. The users will have to re-logon every day now
b. The users will have to manually release and renew their IP addresses to continue working on the network
c. The users will not be affected by the change
d. The users will have to shut their computers down for the duration of the lease to obtain a new lease

Correct answer: c. The users will not be affected by the change

Changing the **DHCP lease time** from **1 week → 1 day** only changes **how long an IP address remains assigned** before renewal is required. However, DHCP clients **automatically renew their IP lease** before it expires. **DHCP renewal process:**

When a client gets an IP address:

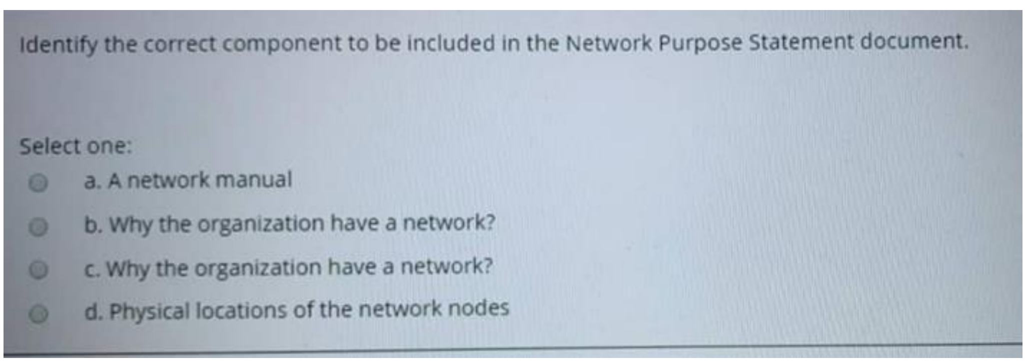
- At **50% of lease time** → client automatically sends a **DHCPREQUEST** to renew the lease.
- If successful → the lease is extended **without user interaction**.

So if the lease becomes **1 day**, the client will try to renew it after **12 hours automatically**.

Therefore: **Users normally will not notice any change.**

Lecture 2 – Network Servers, Baselining & Auditing

Topic: **DHCP Server / DHCP Lease**


Identify the correct component to be included in the Network Purpose Statement document.

Select one:
a. A network manual
b. Why the organization have a network?
c. Why the organization have a network?
d. Physical locations of the network nodes

Correct answer: Why the organization has a network

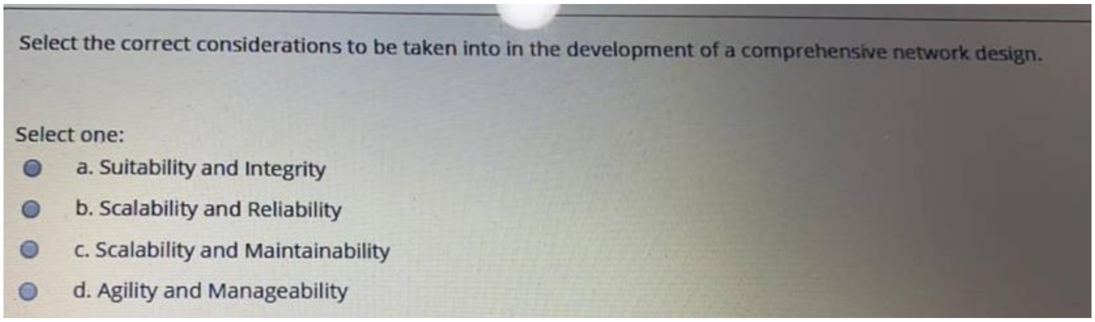
A **Network Purpose Statement** explains **the reason the network exists** and **what the organization wants the network to achieve**.

Typical contents of a network purpose statement include:

- Why the organization needs a network
- Business goals supported by the network
- How the network supports operations

Lecture 1 – Introduction to Network Design & Management

Topic: **Network documentation and planning**



Select the correct considerations to be taken into in the development of a comprehensive network design.

Select one:
- a. Suitability and Integrity
- b. Scalability and Reliability
- c. Scalability and Maintainability
- d. Agility and Manageability

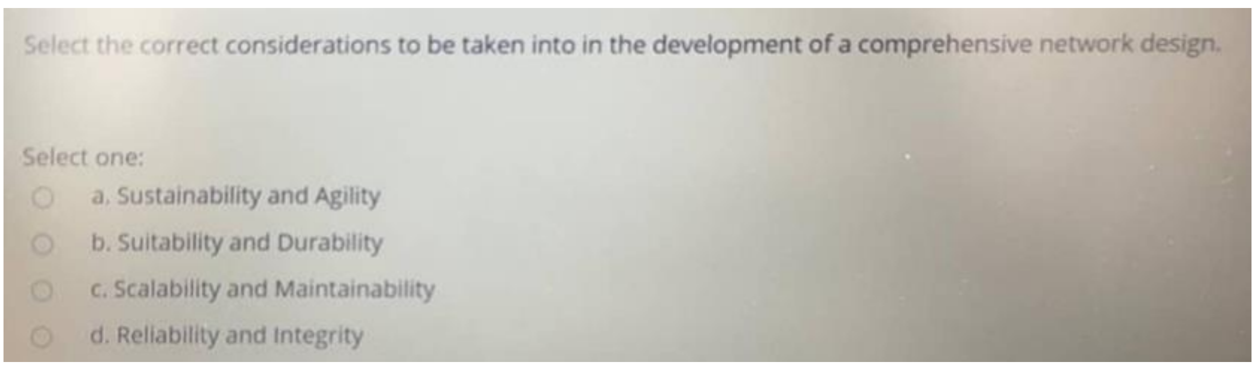
Correct answer: scalability and reliability

When developing a **comprehensive network design**, the considerations are:

- **Suitability**
- **Reliability**
- **Scalability**
- **Durability**

Lecture 1 – Introduction to Network Design & Management

Topic: **Network Design Considerations**



Select the correct considerations to be taken into in the development of a comprehensive network design.

Select one:
- a. Sustainability and Agility
- b. Suitability and Durability
- c. Scalability and Maintainability
- d. Reliability and Integrity

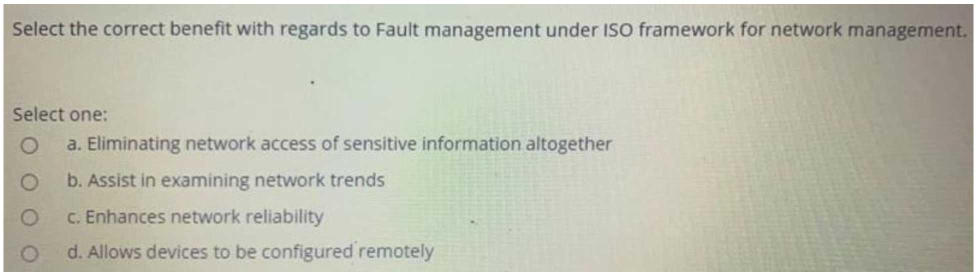
Correct answer: b. Suitability and Durability

comprehensive network design must take into account these four considerations:

- Suitability
- Reliability
- Scalability
- Durability

Lecture 1 – Introduction to Network Design & Management

Topic**: Network Design Considerations**


Select the correct benefit with regards to Fault management under ISO framework for network management.

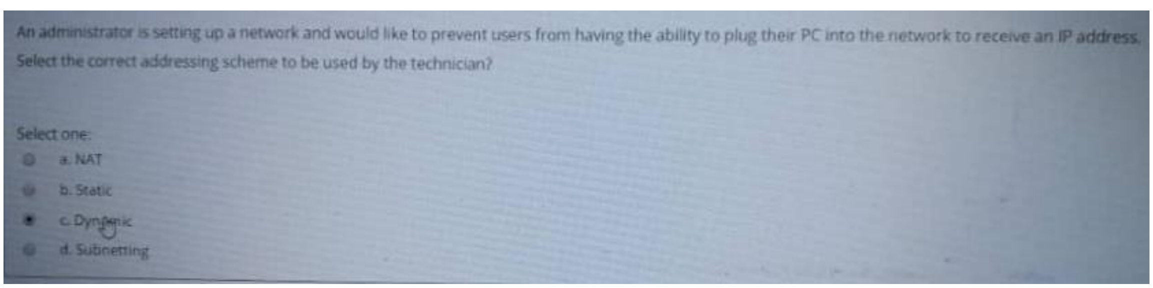
Select one:
- a. Eliminating network access of sensitive information altogether
- b. Assist in examining network trends
- c. Enhances network reliability
- d. Allows devices to be configured remotely

Correct answer: Enhances network reliability

In the **ISO Network Management Framework (FCAPS)**, **Fault Management** is responsible for: detecting network faults / isolating the cause of the fault /correcting or fixing the problem /maintaining normal network operation. The main benefit of fault management is that it **keeps the network operating correctly and reduces downtime**, which directly:

FCAPS Category	Key Benefit
Fault	Improves reliability
Configuration	Remote configuration & inventory
Accounting	Usage tracking & trends
Performance	Consistent service
Security	Protect sensitive data

Lecture 1 – Introduction to Network Design & Management
Topic: **ISO Network Management Framework (FCAPS)**


An administrator is setting up a network and would like to prevent users from having the ability to plug their PC into the network to receive an IP address. Select the correct addressing scheme to be used by the technician?

Select one:
- a. NAT
- b. Static
- c. Dynamic
- d. Subnetting

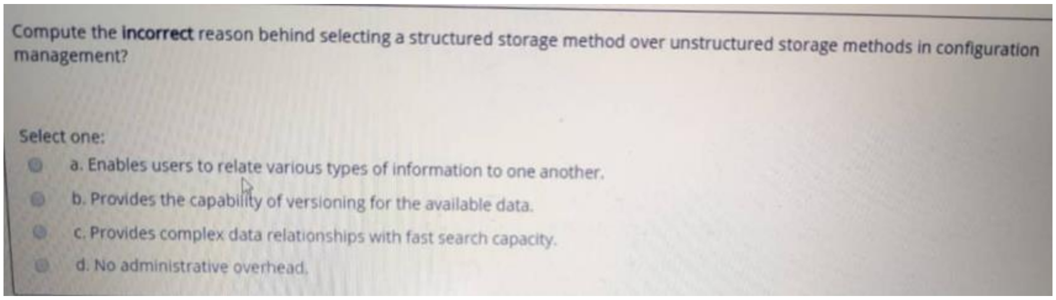
Correct answer: Static

The administrator wants to **prevent users from simply plugging a PC into the network and automatically receiving an IP address**.
This happens when the network uses **dynamic addressing (DHCP)**.
If the administrator instead uses **static IP addressing**, then:

- IP addresses must be **manually configured**
- Only authorized devices with the correct IP configuration can connect
- Unauthorized devices will **not automatically receive an IP address**

Static IP = manual assignment
Dynamic IP = automatic DHCP assignment

Lecture 2 – Network Servers, Baselining & Auditing
Topic: **DHCP / IP Addressing methods**



Compute the **incorrect** reason behind selecting a structured storage method over unstructured storage methods in configuration management?

Select one:
- a. Enables users to relate various types of information to one another.
- b. Provides the capability of versioning for the available data.
- c. Provides complex data relationships with fast search capacity.
- d. No administrative overhead.

Correct answer: d. No administrative overhead

Structured storage (e.g. DBMS) advantages:

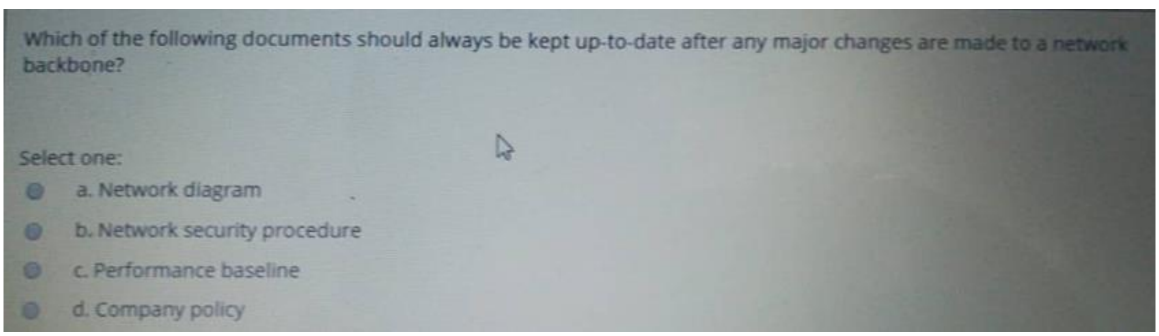
- stores data efficiently
- enables users to relate various types of information to one another
- supports versioning

Structured storage disadvantage:

- administrative overheads
- Requires SQL knowledge

Lecture 1 – Introduction to Network Design & Management

Topic: **Configuration Management → Storing Information**



Which of the following documents should always be kept up-to-date after any major changes are made to a network backbone?

Select one:
- a. Network diagram
- b. Network security procedure
- c. Performance baseline
- d. Company policy

Correct answer: Network Diagram

When a **major change is made to the network backbone**, the document that must be updated immediately is the:

Because the network diagram shows:

- devices
- backbone connections
- topology
- interconnections between network components

If the backbone changes and the diagram is not updated, the documentation becomes inaccurate and troubleshooting/design decisions become harder.

Lecture 2 – Network Servers, Baselining & Auditing

Topic: **Network Mapping / Documentation**

Select the correct goal of network mapping process.

Select one:
- a. To create a complete and accurate picture of the network.
- b. To create a make visualization easier.
- c. To evaluate the security implementations.
- d. To evaluate and baseline network traffic and protocols.

Correct answer: to create a complete and accurate picture of the network

The goal of **network mapping** is to understand the network **inside-out** by creating a detailed description of everything in it.

- Mapping = complete picture
- Baselining = measure performance
- Audit = evaluate current state

Lecture 2 – Network Servers, Baselining & Auditing
Topic: **Network Mapping Definition**

Select the correct components mapped in the Data Link layer under network mapping.

Select one:
- a. Framing and LLC
- b. NIC speed and protocols
- c. IP and Port addressing
- d. Flow Control and Error Control

Correct answer: NIC speed and protocol

In **network mapping**, the **Data Link Layer** is mapped by looking at items related to the **network interface** and link behavior. Because data link layer mapping commonly includes:

- NIC details
- MAC-related behavior
- link protocol information

Mapping question:
Physical layer → cables/devices
Data link layer → NIC
Network layer → IP/routing
Transport layer → ports

Lecture 2 – Network Servers, Baselining & Auditing
Topic: **Network Mapping → Data Link Layer**

Identify who performs accreditation, on domain name registers delegation on whe right to use a domain name.

Select one:

- a. Internet architecture board
- b. Internet research task force
- c. Internet society
- d. Internet corporation for assigned names and numbers

Correct answer: Internet Corporation for Assigned Names and Numbers (ICANN)

ICANN (Internet Corporation for Assigned Names and Numbers) is the organization responsible for:

- coordinating the **global domain name system (DNS)**
- managing **domain name registration**
- accrediting **domain name registrars**
- controlling **IP address allocation**

Organization	Main Role
IAB	Internet architecture oversight
IETF	Internet protocol standards
IRTF	Internet research
ISOC	Internet policy and promotion
ICANN	Domain names & IP address management

An administrator has completed the installation of a new WLAN.
Identify the configuration management documents needs to be updated?

Select one:

- a. Wiring schematics
- b. History logs
- c. Baseline configurations
- d. Logical network diagram

Correct answer: d. Logical network diagram

The administrator has completed installation of a **new WLAN**. That means the **network structure / topology / connectivity** has changed. Because it should now reflect:

- the new wireless segment
- access points
- how clients connect
- changes to the logical network layout

Configuration management is about keeping network information **up to date**, including configuration data and inventory/documentation.

Lecture 1 – Configuration Management

Lecture 2 – Network Mapping / Documentation

Interpret what DNS (Domain Name Server) database contains:

Select one:
- a. None of the mentioned
- b. Hostname-to-address records
- c. Name server records
- d. All of the mentioned
- e. Hostname aliases

Correct answer: d. All of the mentioned

A **DNS database** contains different kinds of **resource records (RRs)**.

DNS records include:

- **A record** → hostname to IP address
- **CNAME record** → hostname alias
- **NS record** → authoritative name server for the domain
- **MX record** → mail server information

So the DNS database does contain:

- **hostname-to-address records**
- **name server records**
- **hostname aliases**

Lecture 2 – Network Servers, Baselining & Auditing

Topic: **DNS Records**

Which of the following protocols should be used for network management?

Select one:
- a. SNMP
- b. SMTP
- c. HTTP
- d. SNTP

Correct answer: a. SNMP

SNMP (Simple Network Management Protocol) is the protocol specifically designed for **network management**.

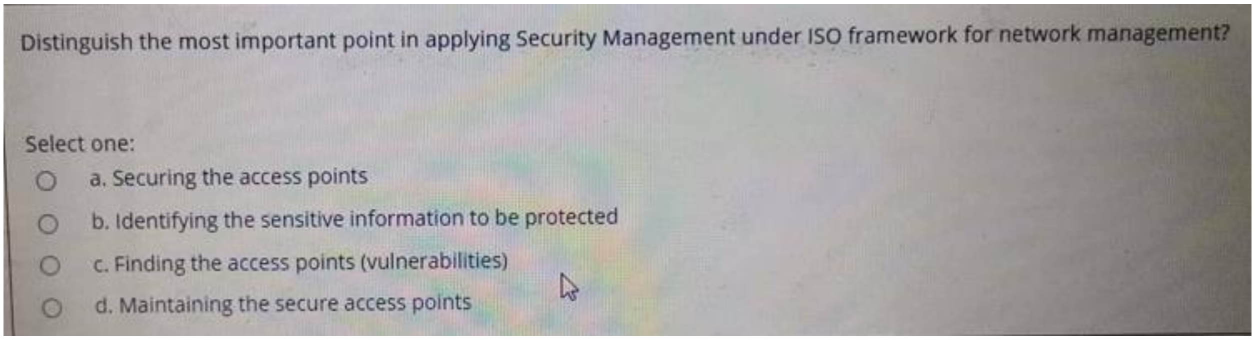
It is used to: monitor network devices / collect performance data / detect faults / manage network configurations. SNMP works using a **manager–agent architecture**, where:

- **Manager** → controls and monitors devices
- **Agent** → software on the network device that reports information

Protocol	Purpose
SNMP	Network management
SMTP	Email transfer
HTTP	Web communication
SNTP	Time synchronization

Lecture 4 – Introduction to SNMP

Topic: **SNMP / Network Management Protocols**



Distinguish the most important point in applying Security Management under ISO framework for network management?

Select one:
- a. Securing the access points
- b. Identifying the sensitive information to be protected
- c. Finding the access points (vulnerabilities)
- d. Maintaining the secure access points

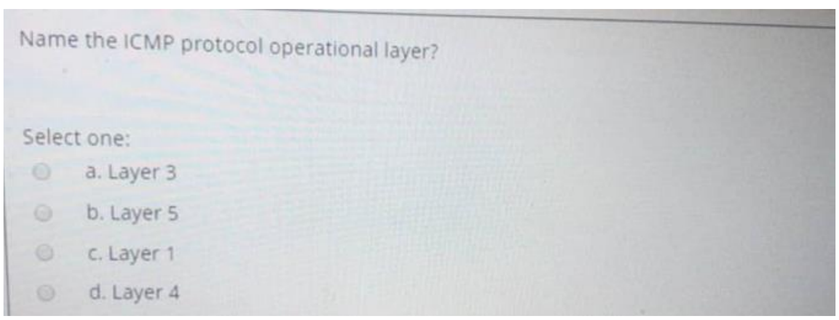
Correct answer: a. Identifying the sensitive information to be protected

Security Management consists of these aspects:

1. **Identifying the sensitive information to be protected**
2. Finding the access points (vulnerabilities)
3. Securing the access points
4. Maintaining the secure access points

Lecture 1 – Introduction to Network Design & Management

Topic: **ISO Network Management Framework → Security Management**



Name the ICMP protocol operational layer?

Select one:
- a. Layer 3
- b. Layer 5
- c. Layer 1
- d. Layer 4

Correct answer: Layer 3 (Network Layer)

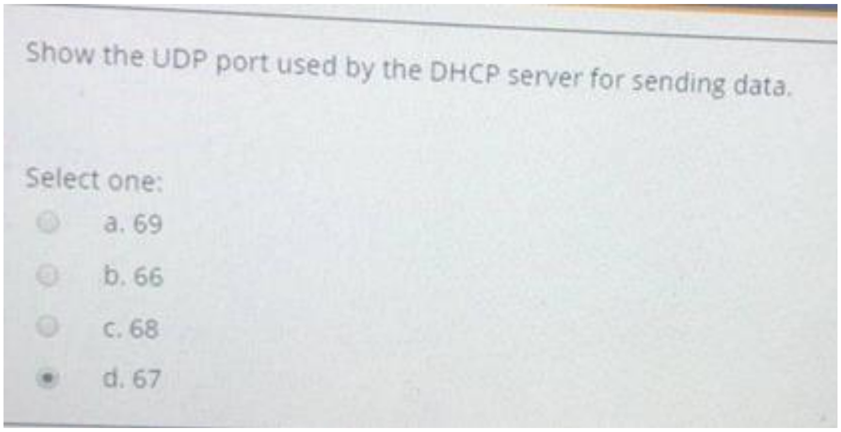
ICMP (Internet Control Message Protocol) operates at the **Network Layer (Layer 3)** of the **OSI model**. ICMP is used for:

- error reporting
- network diagnostics
- sending control messages between network devices

Examples of ICMP usage:

- **Ping** (Echo Request / Echo Reply)
- **Destination Unreachable**
- **Time Exceeded**

Layer	Protocol Examples
Application	HTTP, SMTP, FTP
Transport	TCP, UDP
Network	IP, ICMP
Data Link	Ethernet
Physical	Cables, signals


Show the UDP port used by the DHCP server for sending data.

Select one:
a. 69
b. 66
c. 68
d. 67

Correct answer: 67

DHCP uses two UDP ports:

Device	UDP Port
DHCP Server	**67**
DHCP Client	**68**

The DHCP server sends responses such as:

- **DHCPOFFER**
- **DHCPACK**

from **UDP port 67** to the client.

Protocol	Port
DNS	53
DHCP	67 / 68
HTTP	80
HTTPS	443
SMTP	25
SNMP	161